

```

        face_names.append(name)
        espeak.synth(name)

process_this_frame = not process_this_frame


# Display the results
for (top, right, bottom, left), name in zip(face_locations, face_names):
    # Scale back up face locations since the frame we detected in was scaled
    top *= 4
    right *= 4
    bottom *= 4
    left *= 4

    # Draw a box around the face
    cv2.rectangle(frame, (left, top), (right, bottom), (0, 0, 255), 2)

    # Draw a label with a name below the face
    cv2.rectangle(frame, (left, bottom - 35), (right, bottom), (0, 0, 255), -1)
    font = cv2.FONT_HERSHEY_DUPLEX
    cv2.putText(frame, name, (left + 6, bottom - 6), font, 1.0, (255, 255, 255))

# Display the resulting image
cv2.imshow('Video', frame)

# Hit 'q' on the keyboard to quit!
if cv2.waitKey(1) & 0xFF == ord('q'):
    break

# Release handle to the webcam

```

Fig. 8: The code with `espeak.synth ( )` function

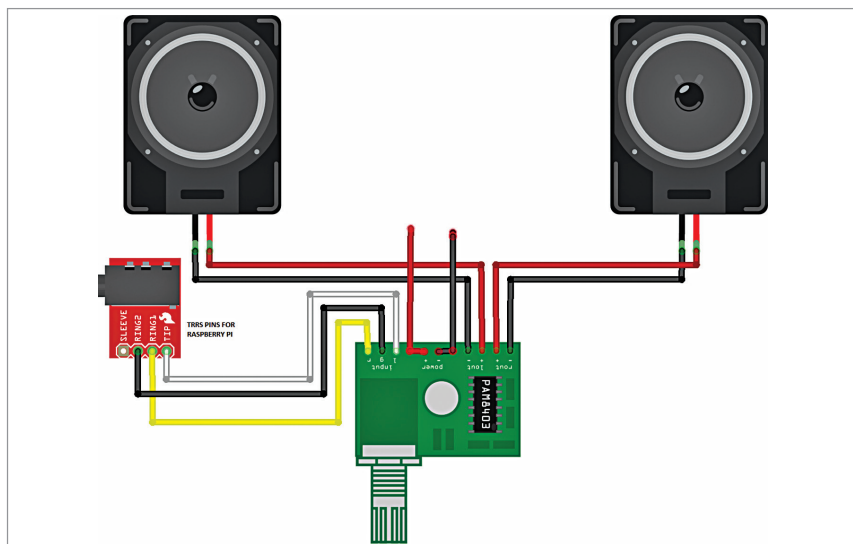


Fig. 9: Connection of audio power amplifier with RPi 4 and speakers

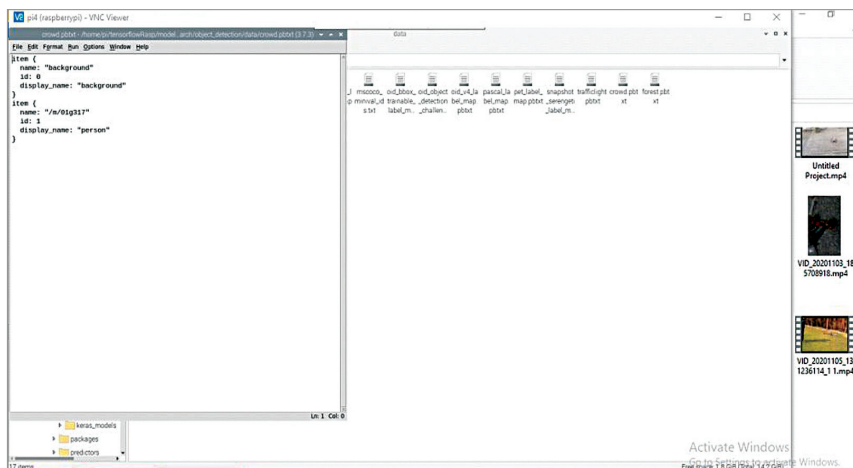


Fig. 10: List of names created in .pbt.txt file

The first step is to update and upgrade RPi 4 operating system and install the required libraries from the terminal by typing the commands given below:

```
sudo apt-get update
sudo apt-get upgrade
sudo nano /etc/dphys-swapfile
```

Then change the line `CONF_SWAPSIZE=100` to `CONF_SWAPSIZE=1024`

```
sudo /etc/init.d/dphys-swapfile stop
sudo /etc/init.d/dphys-swapfile start
```

```
sudo pip3 install opencv
sudo pip3 install numpy
wget https://bootstrap.pypa.io/get-pip.py
pip3 install dlib
```

```
pip3 install tensorflow
```

Next, you can proceed with the cloning of TensorFlow modules, examples and files using the following command:

```
git clone https://github.com/tensorflow/
tensorflow.git
```

After successful cloning, go to the directory→research folder→create a dronecrowd.py Python file. Open the Python3 IDE and import the following modules in the code:

- os
- cv2
- numpy
- tensorflow
- argparse
- sys
- gpiozero
- time

For crowd detection and traffic monitoring, we need to do some setting up before coding. First, we need to create a special .pbtxt file that contains names of the persons/objects to be detected. As our drone is to detect people, so create a crowd.pbtxt file under Tensorflow→models→research→object detection→data folder. Include the display_name as ‘people’ in the crowd.pbtxt file as shown in Fig. 10.

In first part of code (dronecrowd.py), the required libraries are initialised (refer Fig. 11). These include the PiCamera module, which is used to take live video and process it frame-by-frame to detect the crowd during public rallies or gatherings.

Next, define the crowd.pbt.txt file name in the code as shown in Fig. 12.